

Injection scan on the frozen-DINOv2 backbone
(nMSE / free-rollout baseline; red = worse)

[slot] structpos	1.17 $\times$ [†] (0.502)	1.02 $\times$ [†] (0.229)	0.97 $\times$ [†] (0.463)
[slot] +reweight	0.81 $\times$ [†] (0.347)	1.05 $\times$ [†] (0.236)	1.01 $\times$ [†] (0.485)
[slot] +velocity	0.93 $\times$ [†] (0.398)	1.01 $\times$ [†] (0.227)	0.99 $\times$ [†] (0.473)
[probe] probe	1.02 $\times$ [†] (0.434)	1.03 $\times$ [†] (0.231)	1.03 $\times$ [†] (0.493)
[probe] +slot	0.91 $\times$ [†] (0.390)	1.10 $\times$ [†] (0.247)	1.02 $\times$ [†] (0.487)
[dyn] free MLP	0.99 $\times$ [†] (0.423)	1.04 $\times$ [†] (0.234)	1.11 $\times$ [†] (0.530)
[dyn] strict a=g	1.08 $\times$ [†] (0.459)	1.16 $\times$ [†] (0.261)	1.27 $\times$ [†] (0.610)
[cons] consistency	1.00 $\times$ [†] (0.428)	1.01 $\times$ [†] (0.227)	0.96 $\times$ [†] (0.461)
[free] label-free	0.96 $\times$ [†] (0.411)	1.00 $\times$ [†] (0.224)	1.07 $\times$ [†] (0.512)
[free] grounded	1.11 $\times$ (0.473)	1.13 $\times$ (0.254)	1.27 $\times$ (0.607)
	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)